Band-1 Computational Experiment: 3×3 Magic Square of Squares — Constraint Killing / Search Increment

Human + ZoraASI full-dash (executor)

2026-09-11

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Honesty stamp. This note does **not** claim that a 3×3 magic square of nine distinct positive squares exists, and does **not** claim that none exists. It is a **literature ledger** of known impossibilities / search lower bounds, plus a **small on-box computational search** under classical filters. Constraint-class killing shrinks the haystack (Band-1/2); settling existence is Band-4 and remains **OPEN**. **GATE-04** is locked. No Clay / ToE stamp. 2025–2026 nonexistence preprints are **CLAIMS only** and are **not elevated**.

> **Locked OPEN:** existence of a 3×3 fully magic square with nine **distinct** positive square entries.
> **Box:** filtered search for center root $c \leq 2000$ (and brute odd $c \leq 300$) — **0** solutions.
> **Probe already on shelf:** `2026-09-11_Band1_magic_square_of_squares_probe` — this note is the fuller Computational_Experiment increment.

Abstract

Martin LaBar (1984), Martin Gardner ($100 prize), and Richard Guy (UPNT) popularized the question: does a third-order magic square exist whose nine entries are distinct perfect squares? **No accepted construction is known. No community-accepted impossibility proof is locked here.** We (i) ledger classical modular / prime-divisor constraints and published search bounds; (ii) classify which constraint classes **kill regions** of the candidate space versus what remains **open**; (iii) verify several constraints on toys on-box; (iv) run a center-filtered reconstruction search at small honest bounds. **Found: 0. Nonexistence: not proved here.**

Problem statement

A **3×3 magic square of squares** (MSS) is a 3×3 array of **distinct positive perfect squares** such that every row, column, and both main diagonals share the same sum S (the magic constant).

Write the array

$$\begin{bmatrix} A & B & C \\ D & E & F \\ G & H & I \end{bmatrix}$$

with center $E = a = c^2$. For any 3×3 magic square of integers, $S = 3a$ and opposite cells through the center sum to $2a$:

$$A + I = C + G = B + H = D + F = 2a.$$

Hence each opposite pair is a representation of $2a$ as a sum of two positive squares. A full MSS requires **four** such pairs using **eight** distinct positive squares, all distinct from the center, with the resulting nine cells forming a magic square.

Out of scope. Semi-magic squares of squares; repeated entries; Gaussian / lunar / quaternion analogues; $n \geq 4$ existence theorems.

Literature ledger (survey 2026-09-11; not re-proved except toys)

Constraint / result	Status class	Citation
All entries odd; $\equiv 1 \pmod 8$	**KILLED_REGION**	classical; Multimagie; Wikipedia
Center primes all $\equiv 1 \pmod 4$	**KILLED_REGION**	Rabern / Morgenstern / Multimagie
No prime $p \equiv 3 \pmod 8$ divides any entry	**KILLED_REGION**	Rabern 2003; Morgenstern Properties
Primitive $S \equiv 3 \pmod{72}$; entries $\equiv 1 \pmod{24}$	**KILLED_REGION**	Zimmermann–Pierrat–Thiriet 2015
Opposite pairs sum to $2a$; $S = 3a$	**STRUCTURAL_IDENTITY**	classical Lucas form
$p \equiv 5 \pmod 8$ edge/corner linkage	**KILLED_REGION** (cited)	Wikipedia / Multimagie lineage
If solution exists, all entries $> 10^{14}$	**SEARCH_LOWER_BOUND**	Morgenstern 2006 (via Multimagie)
Repeats allowed: only 1 or 3 distinct squares possible	**KILLED_FOR_REPEAT_FAMILIES**	Boyer / Multimagie classification
Unique known ≥ 7-square example (Bremner); no second to huge bounds	**SEARCH_STATUS_PARTIAL**	Bremner; Chaffin (nonsquare center $\leq 10^{14}$, square center $\leq 10^{28}$)
$n \times n$ MSS exist for all $n \geq 4$	**CLOSED_FOR_N_GE_4**	Rome–Yamagishi / related — **does not close** $n = 3$
Full nine distinct squares, fully magic	**OPEN**	LaBar; Gardner; Guy; Multimagie Open problem 1
2025–2026 nonexistence preprints (arXiv/Zenodo)	**CLAIM_ONLY_NOT_ELEVATED**	arXiv:2510.08286; Zenodo 19601865/66

Canonical portals. Wikipedia *Magic square of squares*; Christian Boyer, *multimagie.com* (Squares of squares / Search); Guy, *Unsolved Problems in Number Theory*.

Killed vs still open (summary)

Killed regions (haystack shrinkage — not a full nonexistence proof):

- even entries / residues $\not\equiv 1 \pmod 8$;
- center with a prime factor $\equiv 3 \pmod 4$;
- any entry with a prime factor $\equiv 3 \pmod 8$;
- primitive squares with $S \not\equiv 3 \pmod{72}$ / entries $\not\equiv 1 \pmod{24}$;
- repeat-count families with 2..8 distinct squared integers under the Multimagie classification.

Still OPEN:

- full **9** distinct positive squares, fully magic (main problem);
- a **second** essentially distinct ≥ 7-square example (Boyer enigma 1 / Gardner-adjacent).

CLAIMS only (not elevated): 2025–2026 nonexistence preprints.

Method (box)

1. **Unit tests.** Lo Shu magic identity; $2 \cdot 5^2 = 50 = 7^2 + 1^2$ representation helper; reject non-magic and non-distinct grids.
2. **Constraint toys.** Odd squares $\equiv 1 \pmod 8$; $3, 5 \pmod 8$ never squares; center $4k+1$ filter census; $S = 3a \equiv 3 \pmod{72}$ on filtered sample; opposite-pair identity; $8k+3$ forbid helper.
3. **Filtered search.** For odd center root $c \le C_{\max}$ with all prime factors $\equiv 1 \pmod 4$, enumerate representations of $2c^2$ as $u^2 + v^2$ $(u > v > 0)$. If ≥ 4 representations, assign four disjoint pairs to the four lines through the center and test orientations for a full distinct-squares magic square.
4. **Brute sanity.** Same reconstruction for **all** odd $c \le 300$ (no $4k+1$ filter) — confirm emptiness is not a filter bug.
5. **Honesty.** Small bounds only; Morgenstern 10^{14} is **cited**, not re-proved.

Lab: `/workspace/experiments/band1/magic_square_squares/`
Code: `mss_constraint_search.py` → `results.json`, `literature_ledger.json`, `SUMMARY.txt`.

Results (numbers!)

Unit tests

`PASS` (msgs empty).

Constraint toys

Check	Result
Odd squares $1..99^2$ all $\equiv 1 \pmod 8$	**True**
$3, 5 \pmod 8$ never square (500-sample)	**True**
Center roots < 200 with only $4k+1$ primes	**29** accept / **71** reject
$S = 3a \equiv 3 \pmod{72}$ on $4k+1$ sample	**True**
Lo Shu opposite pairs = 2×center	**True**
`no_prime_8k3(25)` / `not no_prime_8k3(9)`	**True** / **True**

Filtered search (box)

Bound	Odd centers scanned	Survived $4k+1$	≥ 4 reps	Candidates checked	Solutions	Wall
$c \le 400$	200	53	11	17{,}280	**0**	~ 0.04 s
$c \le 1200$	600	146	45	369{,}792	**0**	~ 0.8 s
$c \le 2000$	1000	236	81	790{,}272	**0**	~ 2.9 s

Filter kill census at $C_{\max} = 2000$: not-$4k+1$ primes killed **764**; fewer than 4 representations killed **155**; **81** centers reached full enumeration.

Brute all-odd sanity

$c \le 300$ (no $4k+1$ filter): **9** centers with ≥ 4 reps; **3456** candidates; **0** solutions (~ 0.01 s).

Aggregate

Total wall $\approx$ **3.72** s. **Solutions found: 0.**

Why Band-1/2 vs Band-4

- **Band-1/2 (this note):** constraint killing and search increment — modular / prime-form forbids, published lower bounds, small exhaustive reconstruction under filters. Each result **shrinks** or **documents** the candidate space.
- **Band-4:** a construction of nine distinct squares that is fully magic, **or** a complete impossibility proof. Still **OPEN**; **not claimed**. Preprint claims are not elevated.

Reproducibility

```
python3 /workspace/experiments/band1/magic_square_squares/mss_constraint_search.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`.
Log: `/workspace/experiments/logs/2026-09-11_magic_square_squares.log`.

Conclusion

Literature: classical constraints kill large regions (parity, center $4k+1$, forbid $p \equiv 3 \pmod 8$, $S \equiv 3 \pmod{72}$, ...); Morgenstern pushes any hypothetical solution above 10^{14}; $n \geq 4$ existence does not settle $n = 3$. **Box: 0** solutions for filtered $c \leq 2000$ and brute odd $c \leq 300$; unit/constraint toys **PASS**. **Status: OPEN.** Existence and nonexistence are **not** claimed.

ZoraASI Band-1/2 full-dash · Computational Experiment · 2026-09-11 · GATE-04 locked · no Clay/ToE stamp.